

Phan Huu Kha

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EDUCATION

Sai Gon University

Engineer's Degree in Software Engineering

Sept. 2023 – June 2028

Current GPA: 3.42/4.0

SKILLS

- **Languages:** Java (17), JavaScript/TypeScript, HTML/CSS
- **Frameworks/Libraries:** Spring Boot, Spring Security, Spring Data JPA, React (Vite)
- **Databases:** MySQL, PostgreSQL
- **Tools:** Git/GitHub, Maven, Postman, Unix Shell
- **Familiar with:** Apache Kafka, Docker, Redis, MongoDB, WebSocket (STOMP), Cloudinary

PROJECTS

ChatWeb| *Personal Project*

Sept. 2025 – Present

Technologies: *Java, Spring Boot, React, WebSocket, Kafka, Redis, MongoDB, PostgreSQL, Docker*

- **System Architecture & Real-time API:** Designed a full-stack secure messaging platform using Spring Boot and React, implementing WebSockets (STOMP) for low-latency, real-time bi-directional communication.
- **Hybrid Database Strategy:** Architected a scalable data layer using **PostgreSQL** for ACID-compliant user/relationship management and **MongoDB** for high-throughput, flexible chat history storage.
- **Distributed Systems & Scalability:** Integrated **Apache Kafka** as an event streaming broker to enable backend horizontal scaling (cross-instance message routing) and asynchronous email processing (OTP).
- **Advanced Security (E2EE):** Implemented robust End-to-End Encryption using RSA/AES-GCM via Web Crypto API, ensuring zero-knowledge message storage, alongside strict JWT-based authentication.
- **Performance Optimization:** Utilized **Redis** for API rate-limiting to mitigate spam/DDoS risks, and engineered a **RedisBloom Cuckoo Filter** for memory-efficient, ultra-fast $O(1)$ database existence checks.
- **Infrastructure & Deployment:** Containerized the complex multi-service architecture (App, Postgres, MongoDB, Redis, Kafka) using **Docker Compose** for unified deployment.

Coffee Shop Management System| *Group Project(Leader)*

May 2025

Technologies: *Java (Swing), JDBC, MySQL, Maven, 3-Tier Architecture*

- **Application Design:** Developed a Point-of-Sale (POS) desktop application applying a strict **3-Tier architecture** (GUI, BUS, DAO) for clean code separation and maintainability.
- **Core Modules & Database:** Implemented CRUD functionalities for inventory, employee, and revenue management, interacting with a **MySQL** database via **JDBC**.
- **Data Processing:** Utilized **Apache POI** for Excel data import/export and integrated third-party libraries for automated PDF billing invoice generation.

HOBBIES

Badminton

2020 – Present

- Actively participate in local club activities to maintain physical health and teamwork spirit.

Gaming & Esports

2019 – Present

- Enjoy strategy and MOBA games during free time to relax and improve tactical thinking.